

Gender Differences in Executive Roles based on BoardEx Europe data

DAI Mission - Group E

Emma Kliushnichenko, Mohamed Gomaa, Ahmed Abdelrazek

June 30, 2026

TU Dortmund University - Data Science

- Gender inequality has been widely studied in labour-market outcomes, especially wages and career progression. [1]
- However, inequality is also reflected in access to high-responsibility positions, such as executive and board roles.
- These positions are central to corporate decision-making, strategy, and resource allocation.
- Gender differences in these roles may reflect structural barriers in career progression and leadership selection.
- This is economically relevant because unequal access can affect talent allocation, firm governance, and representation in high-income positions.
- BoardEx Europe allows us to study this question at large scale using detailed role, company, demographic, and education information across 2.2 million organizations based in Europe. [2]

	directorid	female	datestartrole	dateendrole	rolename	...	mba	phd	master	bachelor
1	885377	0.0	1900-01-01	9999-12-31	Director - SD	...	0.0	0.0	1.0	0.0
2	2765991	1.0	1900-01-01	9999-12-31	Executive	...	0.0	0.0	1.0	1.0
3	1321349	0.0	2018-06-01	9000-01-01	Managing Partner	...	0.0	0.0	0.0	1.0

- Six relevant BoardEx tables were merged to create the analytical dataset.
- Role titles from *rolename* and *rolename_clean* were one-hot encoded into role indicators, such as **is_ceo**, **is_cfo**, **is_coo**, and **has_any_executive_role**.
- The same approach was used for education variables, such as **has_mba**, **has_phd**, **has_master**, and **has_bachelor**.
- Heavy gender imbalance: 1,210,252 male vs. 297,309 female observations → **class weighting**.
- Filtered missing gender and missing *datestartrole*: afterwards **1,031,743 rows** remained.
- Replaced placeholders in *dateendrole* with today's date and derived **years_worked**.
- Multiple entries per person, with up to 15 co-occurrences → **inverse directorid weighting**.
- Missing values: median imputation for numeric variables
- Missing categorical values were replaced with the label **Missing**.
- Categories with more than 15 levels were grouped into an **Other** category.

Flag	Rows	%
nationality_missing	667,610	64.71
cccountryname_missing	498,832	48.35
age_missing	404,775	39.23
other_feature_missing	819,502	79.43

Causal Inference

- Treatment *female*, outcome *has_any_executive_role*.
- Assumed **DAG**: adjust for observed confounders (age, age-missing indicator, nationality, HO country, org-type, sector); mediators (education, board status, network) left out → **total** effect.
- **Backdoor adjustment**: propensity-score stratification (deciles) + IPW cross-check.
- **DoWhy** estimation + refutation (random common cause, placebo, 80% subset).
- Missing data: informative *age_missing* indicator + “Missing” category.

Unsupervised Learning

- **K-Means** on individual / organizational context features (+ PCA for visualization); number of clusters via elbow / silhouette.
- Clusters profiled by gender share, sector, board status and executive rate.

Supervised Learning

- Outcome *has_any_executive_role*; executive-defining flags **excluded** (target leakage).
- Models: majority baseline, logistic, lasso logistic, random forest.
- **5-fold stratified CV**; metrics: accuracy, precision, recall, F1, ROC-AUC.
- **Leakage-safe**: imputation + scaling fit *inside* each fold (Pipeline).
- **Class weighting** for the ~80/20 imbalance.
- **Clustered standard errors** on *directorid* (panel structure).

Causal Inference (adjusted gap)

- Naive (unadjusted) gap: -2.5 pp.
- After backdoor / propensity-score adjustment: -4.7 pp (95% CI $[-4.9, -4.5]$).
- Consistent across estimators: DoWhy linear -4.7 , PS-stratification -4.6 , IPW -4.5 pp.
- Adjusting **widens** the gap $\rightarrow$ confounders were masking part of the disparity.

Robustness (refutation)

- Random common cause $\rightarrow$ estimate unchanged.
- Placebo treatment $\rightarrow$ effect ≈ 0 .
- 80% data subset $\rightarrow$ estimate unchanged.

Supervised Learning

Model	Accuracy	F1	ROC-AUC
Baseline (majority)	0.67	0.00	0.50
Logistic	0.77	0.73	0.86
Lasso Logistic	0.77	0.73	0.86
Random Forest	0.77	0.74	0.86

- *female*: odds ratio **0.83** ($\sim 17\%$ lower odds), conditional on all other features.
- Lasso **keeps** *female* $\rightarrow$ independent signal.
- Clustered SEs (on *directorid*): gender stays significant ($p < 0.001$).
- Top drivers: board/role context (NED, partner, board membership), then age.

Synthesis and Limitations

Synthesis

- The causal block estimated the adjusted gender gap: the estimated gap for women became larger after adjustment.
- The supervised learning block then checked whether this negative gender pattern also appeared in prediction. Logistic regression showed a negative association between *female* and executive access, and lasso kept *female* non-zero.
- The random forest helped show which variables influenced prediction: role and board variables were more important than other variables, showing that existing seniority was important for executive access.
- The unsupervised learning block finally added context by showing that these role records were not homogeneous. The data were dominated by one large mixed cluster, while smaller clusters captured more specific role profiles, such as a non-executive board group and a Health/Sweden group with the highest female share and executive rate.
- Overall, the research answered the question of which individual characteristics influenced the probability of holding a top position in a company, and whether gender remained associated with this probability after accounting for these factors. Gender still appeared to be an important indicator even after several factors were included, but the results should be interpreted as evidence of an existing gender gap, not as a complete explanation of it.

Limitations

- The data were observational, so the analysis could not fully prove that gender itself caused lower executive access.
- The country distribution was uneven: Germany and France dominated the sample, so patterns from smaller countries may have been harder to detect.
- Cluster labels were not fed back into the supervised models to test predictive gains; the gender gap was, however, examined within clusters (see backup) and is concentrated in the largest segment.
- K-means and plotting with PCA were limited for many categorical and one-hot encoded variables. We should have tried to include more continuous variables, for example, *years_worked*.

References

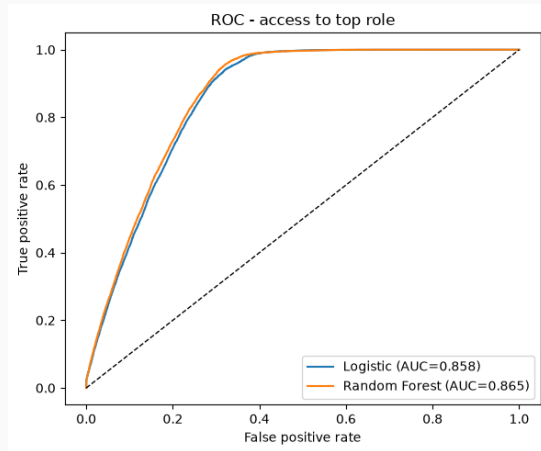
- [1] The Nobel Prize. *The Prize in Economic Sciences 2023: Claudia Goldin*.
<https://www.nobelprize.org/prizes/economic-sciences/2023/press-release/>.
Accessed: 2026-06-30. 2023.

- [2] Wharton Research Data Services. *BoardEx Europe*.
<https://wrds-www.wharton.upenn.edu/pages/get-data/boardex/boardex-europe/>.
Accessed: 2026-06-29. 2026.

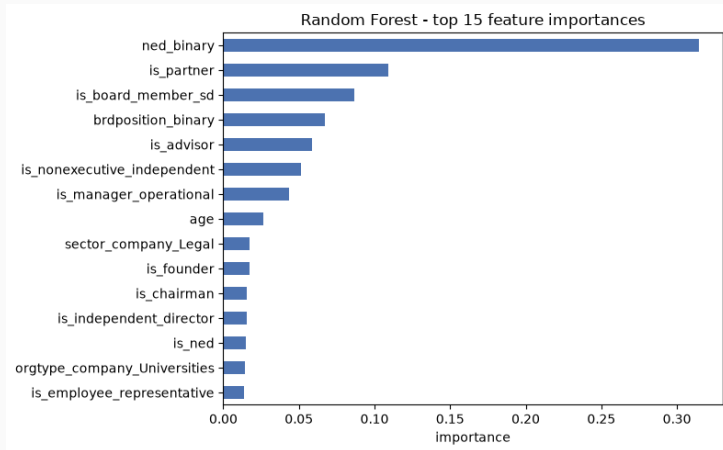
Model	Acc.	Prec.	Recall	F1	AUC
Baseline (majority)	0.67	0.00	0.00	0.00	0.50
Logistic	0.77	0.59	0.94	0.73	0.86
Lasso Logistic	0.77	0.59	0.94	0.73	0.86
Random Forest	0.77	0.60	0.95	0.74	0.86

Leakage-safe (in-fold imputation), class-weighted. Models agree $\Rightarrow$ largely linear.

Backup — ROC Curves



Backup — Random Forest Feature Importance



Backup — Causal: Propensity-Score Stratification

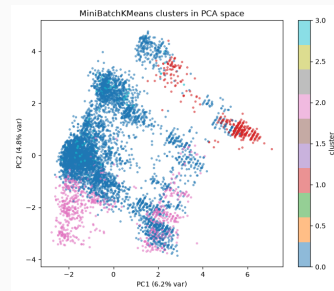
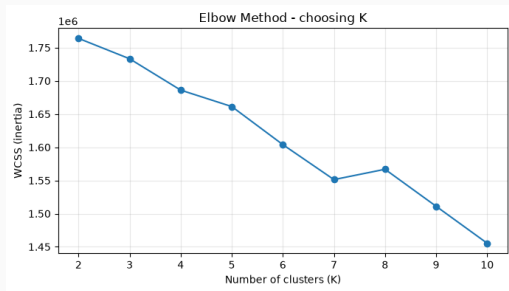
Stratum	n_F	n_M	exec_F	exec_M	gap
0	8408	95022	0.157	0.188	-0.030
1	14748	88177	0.223	0.289	-0.066
2	18151	85018	0.259	0.321	-0.062
3	22923	94763	0.314	0.368	-0.054
4	19068	70667	0.318	0.375	-0.058
5	26963	87854	0.318	0.361	-0.044
6	21961	70015	0.338	0.369	-0.031
7	27576	76808	0.329	0.367	-0.038
8	28618	71844	0.342	0.386	-0.044
9	34729	68426	0.349	0.394	-0.045

Negative gap in every stratum on common support; weighted ATE = -4.7 pp.

Cluster	<i>n</i>	%	Female	Exec rate	Top sector	Top country
0	850,248	82.4	0.50	0.42	Missing	Other
1	45,351	4.4	0.45	0.00	Missing	UK–England
2	114,709	11.1	0.48	0.13	Missing	UK–England
3	21,435	2.1	0.54	0.46	Health	Sweden

Cluster 3 (Health/Sweden): highest female share and executive rate; cluster 1: non-executive board group (exec rate 0).

Backup — Unsupervised: Elbow & Cluster Map



Left: WCSS elbow ($K = 4$ chosen). Right: clusters in PCA space (PC1 6.2%, PC2 4.8%).

Backup — Heterogeneous Gender Gap

Cluster	n	exec_F	exec_M	gap (F–M)
0 (large mixed)	850,248	0.341	0.378	–3.7 pp
1 (non-exec board)	45,351	0.000	0.000	≈ 0
2	114,709	0.176	0.163	+1.3 pp
3 (Health/Sweden)	21,435	0.387	0.408	–2.2 pp

The penalty is concentrated in the large mixed cluster (–3.7 pp) and mildly reverses in cluster 2 $\Rightarrow$ the gender gap is **not uniform** across corporate environments.